How-To Library

Informatica

Installing Secure@Source on a Kerberized Hortonworks 2.6 External Cluster

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Abstract

This guide describes the installation process for Secure@Source in a Linux environment, on a Kerberos-enabled Hortonworks 2.6 external cluster.

Supported Versions

- Secure@Source 4.0

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Overview

The Secure@Source installer is designed to create the domain and create and configure the services necessary for Secure@Source. However, the installation fails when you attempt to install Secure@Source on a Kerberos-enabled Hortonworks 2.6 external cluster. The installation fails because the Slider is not able to push the applications into Apache Hadoop YARN. Consequently, the Catalog Service fails to come up.

To resolve this issue and complete the Secure@Source installation, perform the following steps:

1. Verify the prerequisites.
2. Run the Secure@Source installer.
3. Apply the Emergency Bug Fix (EBF) 9873.
4. Verify that the application services are enabled.
5. Create the Secure@Source Service.
6. Configure the Secure@Source Service.

Step 1. Verify Prerequisites

1a. Prerequisites for the Secure@Source Server and Informatica Services Node

Refer to the H2L: Secure@Source Quick Start Deployment Guide: Embedded Hadoop Cluster, Step 1.

1b. Prerequisites for the External Cluster

OpenSSL

Openssl must be version 1.0.1 or above.

Java 8

The external cluster must have Java 8.

Configure YARN memory and cores

Example:

- yarn.scheduler.maximum-allocation-mb = 20 GB
- yarn.nodemanager.resource.memory-mb = 64 GB
- yarn.scheduler.maximum-allocation-vcores = 16
- yarn.nodemanager.resource.cpu-vcores = 16

Create HDFS directories

Create the following directories in HDFS. Enterprise Information Catalog will use the directories as data directories for the Catalog Service.

```
 /Informatica/LDM/<service cluster name>
/Informatica/LDM/<service cluster name>/service-logs/
/user/<username>
```

Where `<service cluster name>` is the name of the service cluster that you need to enter when you create the Catalog Service and <username> is the user name of the Informatica domain user.

Make the Informatica domain user the owner of the `/Informatica/LDM/<ServiceClusterName>` and `/user/<username>` directories.

Change the group of `/Informatica/LDM/<Service cluster name>/service-logs` to `hadoop`.

The default service cluster name is <DomainName>_<CatalogServiceName>. Use the default only if Kerberos is not enabled.

1c. Pre-installation Steps for Kerberos Authentication

Perform the following steps If the external cluster uses Kerberos authentication.

Note that Service Cluster Name is the name of the service cluster. This name is entered during installation and must match the principal user created in the KDC. Informatica domain user is the OS user that installs, owns, and executes the Secure@Source software.

Configure the Key Distribution Center (KDC) hostname and IP address

You must configure the Key Distribution Center (KDC) hostname and IP address on all cluster nodes and domain machines in the `/etc/hosts`.

Ensure that the `krb5.conf` file is located in all cluster nodes and domain machines under the `/etc` directory.

Create LDAP user

Create a user in the LDAP security domain where <username> matches the service cluster name.

Create Principal users

Create the following principal users in the LDAP security domain where <username> is the service cluster name.

```
<username>@KERBEROSDOMAIN.COM
<username>/<hostname>@KERBEROSDOMAIN.COM
```

Note: Create the principal user for all hosts in the cluster.

```
HTTP/<hostname>@KERBEROSDOMAIN.COM
```

Note: The user should already exist. If not, create the user for all hosts in the cluster.

Example:

```
kadmin.local -q 'addprinc -pw hadoop infa@EXAMPLE.COM'
 kadmin.local -q 'addprinc -pw hadoop
infa/cdh2.example.com@EXAMPLE.COM'
```

Create and merge the keytab file

Create a keytab file with above principals and merge this keytab file with Hortonworks `/etc/security/keytabs/spnego.service.keytab` files on Hadoop node(s) as this keytab contains the HTTP principal.

Copy keytab to the Secure@Source Server

Copy the keytab created above to a directory on the Secure@Source server.

udp_preference_limit

Add the udp_preference_limit = 1 to `/etc/krb5.conf`

Set the udp_preference_limit on both the Domain and Hadoop machines.

Set up the udp_preference_limit parameter in the `krb5.conf` Kerberos configuration file to 1. This parameter determines the protocol that Kerberos uses when it sends a message to the KDC. Set `udp_preference_limit` to 1 to always use TCP.

The Informatica domain supports only the TCP protocol. If the `udp_preference_limit` parameter is set to any other value, the Informatica domain might shut down unexpectedly.

Create the elasticsearch directory

Create `/var/lib/hadoop-yarn/<SecureAtSourceServiceName>_elasticsearch` directory.

The owner of this folder must be the same as the Principal User. The Principal User must be the same as the Service Cluster Name. The Secure@Source service name in the directory name must be in lowercase.

Example:

```
mkdir /var/lib/hadoop-yarn/infa_sats_elasticsearch
chown infa:infa /var/lib/hadoop-yarn/infa_sats_elasticsearch
```

Note the Kerberos Information for the Installer

Make a note of the values for the following Kerberos prompts you will have to respond to when you run the Secure@Source installer:

- Is Cluster Kerberos-enabled? **2**
- Select the Hadoop Distribution Type and provide the cluster details.
- Service Cluster Name: Must be the same as the Principal user.
- HDFS Service Principal Name: The HDFS Principal that you configured during the Kerberos setup of Hortonworks. The default setup of Hortonworks Kerberos will have `nn/_HOST@KERBEROSDOMAIN.COM` as Principal.
- Yarn Service Principal Name: The Yarn Principal that you configured during the Kerberos setup of Hortonworks. The default setup of Hortonworks Kerberos will have `rm/_HOST@KERBEROSDOMAIN.COM` as Principal.
- KDC Domain Name: Specify the KDC Domain name for example: HADOOP.COM

- Keytab Location: Specify the keytab file location with filename that you created in the *Create and merge the keytab file* section above.
- Kerberos Configuration File location: Specify the location with filename of the Kerberos configuration file. Example: `etc/krb5.conf`

Step 2. Run the Secure@Source Installer

Run the Secure@Source installer until you get the message that the installation failed.

1. Download the Secure@Source installer and extract the installation files.
2. On a shell command line, run the install.sh file from the installation directory.
3. Specify the appropriate choices and enter the values for the prompts on each panel.
4. On the **Cluster Type Selection** panel, at the Hadoop cluster type prompt, choose **2** to indicate that you want to deploy Secure@Source on an external Hadoop distribution on Hortonworks version 2.6.
5. To specify that the cluster is Kerberos-enabled, select **2**.
6. Enter values for the following parameters:

Property	Description
Gateway User	Username for the Apache Ambari server.
Informatica Cluster Service Name	Name of the Hadoop service for the internal cluster.
Informatica Cluster Service Port	Port number for the Hadoop service.
Informatica Hadoop Cluster Gateway Host	Host where the Apache Ambari server runs.
Informatica Hadoop Cluster Nodes	Hosts where the Apache Ambari agents run.
Informatica Hadoop Cluster Gateway Port	Web port for the Apache Ambari server.
Informatica Hadoop Service HTTPS Port	HTTPS port number for the Hadoop Service.
Catalog Service Name	Name of the catalog service.
Catalog Service Port	Port number of the catalog service.
Hadoop Trust Store File	Path of the Hadoop truststore file.

7. Continue specifying choices and entering values.
8. If you use the default SSL, Catalog Service startup fails with the following error:

```
java.util.concurrent.ExecutionException:
com.infa.products.ldm.http.utils.RestClientException: [RestClientException_00004] Unable
to connect to Yarn Resource Manager due to the following error
[[javax.net.ssl.SSLHandshakeException: sun.security.validator.ValidatorException: No
trusted certificate found]]. Please verify Yarn Resource Manager is running
```

To resolve, perform the following tasks:

a. Import each hadoop node certificate in to the Domain truststore.

b. Import the Catalog Service keystore certificate in to the Yarn truststore.

9. The installation fails with the following error:

```
java.lang.RuntimeException: Entry "Client" not found; JAAS config = ;
java.security.auth.login.config=(undefined)
```

10. Exit the installer.

Step 3. Apply the Secure@Source EBF

Install the Informatica EBF-9873 to circumvent the issue that occurs when Secure@Source is installed on Hortonworks 2.6.

1. Shutdown the domain and all nodes.
   a. Navigate to the directory where the `./infaservice.sh` file is located:

      ```
      cd $INFA_HOME/server/tomcat/bin
      ```

   b. Enter: `./infaservice.sh shutdown`
2. Delete `$INFA HOME/tomcat/temp/<Catalog Service name>`.

   For example, if the Catalog Service name is **CS**, enter: `rm -rf $INFA HOME/tomcat/temp/CS`
3. Download the EBF-9873.
4. Extract (untar) the EBF Installer.
5. The EBF Installer includes the following files:
   - `Input.properties`
   - `installEBF.sh`
6. Open the `Input.properties` file in edit mode and set DEST_DIR=`<10.2.x installation dir>`.

   Example: DEST_DIR=`/home/infauser/10.2.0`
7. Install the EBF. Run: `./installEBF.sh`

   A log file is generated in the installation directory <10.2.x installation dir>. The error log is generated in the same folder where the EBF installer is extracted with the file name: `EBF_<EBFID>_Error.log`. Example: `EBF_EBF252831_Error.log`
8. Startup the domain and all nodes.
   a. Navigate to the directory where the `./infaservice.sh` file is located:

      ```
      cd $INFA_HOME/server/tomcat/bin
      ```

   b. Enter: `./infaservice.sh startup`

Step 4. Verify that the Application Services are Enabled

Log in to the Informatica Administrator tool and confirm that the application services are enabled.

1. In the Address field of a browser, enter the URL for the Administrator tool.
   - If the Administrator tool is not configured to use a secure connection, enter the following URL:

     ```
     http://<fully qualified hostname>:<http port>/administrator/
     ```

   - If the Administrator tool is configured to use a secure connection, enter the following URL:

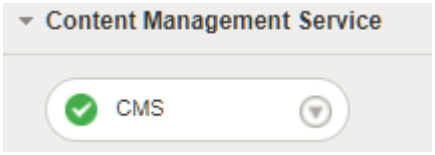
     ```
     https://<fully qualified hostname>:<https port>/administrator/
     ```

   Host name and port in the URL represent the host name and port number of the master gateway node.
2. Enter the user name and password and then click **Login**.
3. Click the **Manage** tab.

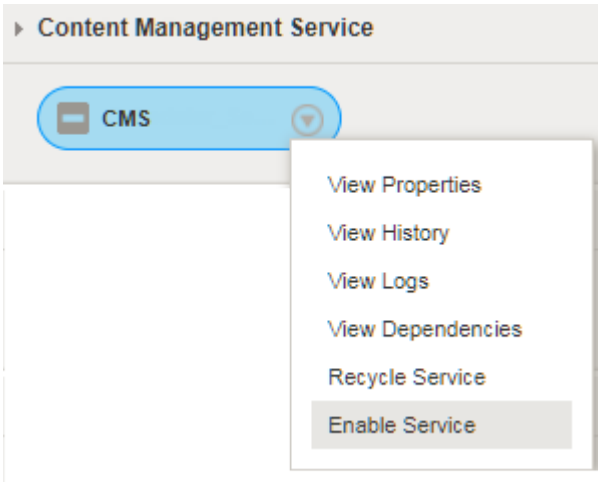
4. Click the **Domain** view.

   A list of the services and state appears.
5. Verify that the following services are enabled.
   - Catalog Service
   - Content Management Service
   - Data Integration Service
   - Informatica Cluster Service
   - Model Repository Service

   Services that are enabled will have a green checkmark.



6. If a service is not enabled, click the **Actions** menu next to the name of the service and select **Enable Service**.



Step 5. Create the Secure@Source Service

From the Administrator tool, create the Secure@Source Service.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Click the domain name in the Domain Navigator pane.
4. Click the **Actions** menu in the Domain Navigator pane and select **New** > **Secure@Source Service**.

   The **New Secure@Source Service** dialog box appears.

5. On the **New Secure@Source Service - Step 1 of 4** page, enter the following properties:

Property	Description
Name	Name of the service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [You cannot change the name of the service after you create it.
Description	Description of the service. The description cannot exceed 765 characters.
Location	Domain and folder where the service is created. Click **Browse** to choose a different folder. You can move the service after you create it.
License	License object that allows use of the service. You cannot edit this property.
Node	Node on which the service runs.

6. Click **Next**.

   The **New Secure@Source Service - Step 2 of 4** page appears.

7. Enter the following properties for the Secure@Source repository database:

Property	Description
Database Type	The type of the repository database.
URL	The JDBC connection string used to connect to the Secure@Source repository database. Use the following JDBC connect string syntax for each supported database: - IBM DB2. `jdbc:informatica:db2://<host_name>:<port_number>;DatabaseName=<database_name>;BatchPerformanceWorkaround=true;DynamicSections=3000` - Microsoft SQL Server that uses the default instance. `jdbc:informatica:sqlserver://<host_name>:<port_number>;DatabaseName=<database_name>;SnapshotSerializable=true` - Microsoft SQL Server that uses a named instance. `jdbc:informatica:sqlserver://<host_name>\<named_instance_name>;DatabaseName=<database_name>;SnapshotSerializable=true` - Oracle. `jdbc:informatica:oracle://<host_name>:<port_number>;SID=<database_name>;MaxPooledStatements=20;CatalogOptions=0;BatchPerformanceWorkaround=true`
Secure JDBC Parameters	The secure database parameters.
User Name	The database user name for the repository.
Password	Repository database password for the database user.

Property	Description
Schema	The schema name for a particular database.
Tablespace	The tablespace name for a particular database. For a multi-partition IBM DB2 database, the tablespace must span a single node and a single partition.

8. Click **Test Connection** to verify that you can connect to the database.
9. Select the following option to create content for the Secure@Source repository:
   - **No content exists under specified connection string. Create new content.**

   Note: The Content Management Service must be running to create content for the Secure@Source Service. If you create content for the Secure@Source Service after you create the service, you must first disable the service. If you do not disable the service, you cannot create content.
10. Select **Enable the Secure@Source Service** to automatically enable the service after you create the service. If disabled, you must manually enable the service after you create the service.
11. Click **Next**.

    The **New Secure@Source Service - Step 3 of 4** page appears.
12. Enter the following properties for the associated application services:

Property	Description
Catalog Service Name	Name of the Catalog Service that you want to associate with the Secure@Source Service. The Catalog Service is an application service that runs Live Data Map in the Informatica domain.
Persistent Masking Service Name	Name of the Persistent Masking Service that you want to associate with the Secure@Source Service.
User Name	User name that the Secure@Source Service can use to access the Catalog Service and Persistent Masking Service.
Password	Password for the Catalog Service and Persistent Masking Service user.

13. Click **Next**.

    The **New Secure@Source Service - Step 4 of 4** page appears.
14. Enter the following HTTP configuration and SSL configuration properties:

Property	Description
HTTP Port	Port number on which the Secure@Source application runs. Default is 6200.
Enable Secure Communication	Enables secure communication for the Secure@Source Service in the domain.
HTTPS Port	Port number to use for a secure connection to the service. Use a different port number than the HTTP port number.

Property	Description
Keystore File	Directory that contains the keystore file that has the digital certificates.
Keystore Password	Password for the keystore file.

The domain creates the Secure@Source Service, creates content for the Secure@Source repository in the specified database, and enables the service.

15. Click **Finish**.

Step 6. Configure the Secure@Source Service

From the Administrator tool, configure the following types of properties:

- Secure@Source Service Properties
- Secure@Source Service Process Properties

Step 6a. Configure the Secure@Source Service Properties

The Secure@Source Service properties are organized in the following sections:

- General
- Secure@Source repository
- Associated services
- User activity configuration
- Advanced
- Email server configuration

General Properties for the Secure@Source Service

The general properties of a Secure@Source Service include name, license, and node assignment.

You can configure the following general properties for the service:

Name

Name of the service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters:

` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [

You cannot change the name of the service after you create it.

Description

Description of the service. The description cannot exceed 765 characters.

License

License object that allows use of the service.

You cannot edit this property.

Node

Node on which the service runs.

Secure@Source Repository Properties for the Secure@Source Service

You can configure the following Secure@Source repository properties for the Secure@Source Service:

Database Type

The type of the repository database.

URL

The JDBC connection string used to connect to the Secure@Source repository database.

Use the following JDBC connect string syntax for each supported database:

- IBM DB2. `jdbc:informatica:db2://<host_name>:<port_number>;DatabaseName=<database_name>;BatchPerformanceWorkaround=true;DynamicSections=3000`
- Microsoft SQL Server that uses the default instance. `jdbc:informatica:sqlserver://<host_name>:<port_number>;DatabaseName=<database_name>;SnapshotSerializable=true`
- Microsoft SQL Server that uses a named instance. `jdbc:informatica:sqlserver://<host_name>\<named_instance_name>;DatabaseName=<database_name>;SnapshotSerializable=true`
- Oracle. `jdbc:informatica:oracle://<host_name>:<port_number>;SID=<database_name>;MaxPooledStatements=20;CatalogOptions=0;BatchPerformanceWorkaround=true`

Secure JDBC Parameters

If the Secure@Source repository database is secured with the SSL protocol, you must enter the secure database parameters.

Enter the parameters as `name=value` pairs separated by semicolon characters (;). For example:

```
param1=value1;param2=value2
```

User Name

The database user name for the repository.

Password

Repository database password for the database user.

Schema

The schema name for a particular database.

Tablespace

The tablespace name for a particular database. For a multi-partition IBM DB2 database, the tablespace must span a single node and a single partition.

Associated Services for the Secure@Source Service

You can configure the following associated services properties for the Secure@Source Service:

Catalog Service Name

Name of the Catalog Service that you want to associate with the Secure@Source Service. The Catalog Service is an application service that runs Live Data Map in the Informatica domain. Select a service from the drop-down menu.

Persistent Masking Service Name

Name of the Persistent Masking Service that you want to associate with the Secure@Source Service. Select a service from the drop-down menu.

User Name

User name that the Secure@Source Service can use to access the Catalog Service and Persistent Masking Service.

Password

Password for the Catalog Service and Persistent Masking Service user.

User Activity Configuration Properties for the Secure@Source Service

You can configure the following user activity properties for the Secure@Source Service:

Enable User Activity

When enabled, ensures user activity data is streamed to Secure@Source.

Default is False.

Note: Although the TCP protocol ensures reliable transmission at the network layer, when messages are sent to the TCP listener at a very high rate, the application layer buffer on the listener might fill up with messages faster than the listener can process. In this event, some incoming messages might be dropped at the listener and not streamed to Secure@Source.

Event Details Retention Period

Determines the number of days to retain user activity details and anomalies in the user activity store. The Secure@Source Service runs a daily retention job that purges expired data from the user activity store.

Advanced Properties for the Secure@Source Service

You can configure the following advanced property for the Secure@Source Service:

Minimum Conformance Percentage

Specify the minimum percentage of values in a field that must match the data domain data match condition for Secure@Source to identify the field as sensitive.

The default value is 80%.

User Activity Application Port Range

Specify the port range for user-activity applications. The range must include at least 10 ports. Enter the minimum and maximum port numbers in the range separated by an hyphen.

Default is 40000 - 50000.

Email Server Configuration Properties for the Secure@Source Service

You can configure the following email server configuration properties for the Secure@Source Service:

Server Host Name

The SMTP outbound mail server host name. For example, enter the Microsoft Exchange Server for Microsoft Outlook.

Server Port

Port number used by the outbound SMTP mail server. Valid values are from 1 to 65535.

User Name

User name for authentication, if required by the outbound SMTP mail server.

Password

Password for authentication, if required by the outbound SMTP mail server.

Authentication Enabled

Indicates that the SMTP server is enabled for authentication. If true, the outbound mail server requires a user name and password.

Use Security

Indicates that the SMTP server uses SSL or TLS protocol.

Security Protocol

The SSL or TLS port number for the SMTP server port property.

Sender Email Address

Email address that the Secure@Source Service uses in the From field when the service sends notification emails.

Step 6b. Configure the Secure@Source Service Process Properties

The Secure@Source Service process properties are organized in the following sections:

- Security Configuration
- Environment Variables
- Logger Options
- Advanced Process Configuration

Security Configuration for the Secure@Source Service Process

You can configure the following security properties for the Secure@Source Service process:

HTTP Port

Unique HTTP port number to use for the Secure@Source Service.

Default is 6200.

Enable Secure Communication

When enabled, ensures secure and encrypted communication with the Secure@Source Service. When you enable secure communication, you must provide a HTTPS port and the keystore file and password.

Note: After you enable secure communication, you can no longer switch to HTTP mode.

Default is true.

HTTPS Port

Unique HTTPS port number to use for a secure connection to the Secure@Source Service. Use a different port number than the HTTP port number. Required if you select Enable Secure Communication.

Keystore File

Path and file name of the keystore file that contains the private or public key pairs and associated certificates. Required if you select Enable Secure Communication.

You can create a keystore file with a keytool. keytool is a utility that generates and stores private or public key pairs and associated certificates in a keystore file. You can use the self-signed certificate or use a certificate signed by a certificate authority.

Keystore Password

Plain-text password for the keystore file.

Environment Variables for the Secure@Source Service Process

You can configure the following environment variable property for the Secure@Source Service process:

Environment Variable

Environment variables defined for the Secure@Source Service process.

Logger Options for the Secure@Source Service Process

You can configure the following logger options for the Secure@Source Service process:

Logging Level

The severity level for Secure@Source logs. Valid values are off, error, info, debug, trace, all. Default is info.

Maximum Log File Size

The maximum size for a log file in KB, MB, or GB. Configure a maximum size to enable log file rollover by file size. When the log file reaches the maximum size, the Secure@Source Service creates a new log file. The Secure@Source Service can create up to five backup log files.

Enter a number, followed by a unit of measurement. You use the following units of measurement:

- K. Kilobytes
- M. Megabytes
- G. Gigabytes

If you do not enter a unit of measurement, the Secure@Source Service uses megabytes.

Default is 200M. Do not use a space after the number.

Advanced Logger Options

Custom log4j.logger properties.

For example, you might set the following property:

```
log4j.logger.org.apache.jasper.servlet.jspservlet=INFO
```

Advanced Properties for the Secure@Source Service Process

You can configure the following advanced properties for the Secure@Source Service process:

Number of Parallel Jobs

The maximum number of parallel jobs. Default is 100.

Additional JVM Options

Custom Java Virtual Machine (JVM) options.

Note: Set the heap size in the Maximum Heap Size property, not in the Additional JVM Options property.

You can add the following properties:

-DdisableBlazeMode

Determines the engine that a Hadoop scan job uses to run mappings. By default, a Hadoop scan job uses the Blaze engine to run mappings.

To enable a Hadoop scan job to use the Hive engine to run mappings, add the property in the following format:

```
-DdisableBlazeMode=true
```

-DIncludeUnknownConnections

Determines if an Informatica PowerCenter scan job imports data stores that contain an unsupported data store type in Secure@Source. For example, a scan job can import a data store that connects to Siebel. Siebel is not a supported data store type. When the scan job imports an unsupported data store, the scan job creates a data store with data store type JDBC. You must complete the data store properties and configure JDBC connectivity to the source. After you scan the imported data stores, the scan job can identify the proliferation of the imported data stores.

To enable an Informatica PowerCenter scan job to import unsupported data stores, add the -DIncludeUnknownConnections property in the following format:

```
-DIncludeUnknownConnections=Y
```

-DSATS_THREAD_COUNT

Specifies the number of the threads that the Secure@Source Service uses for the Evaluate Classification Policies job and the Collect Row Count and Evaluate Classification Policies scan job steps.

To change the number of threads, add the -DSATS_THREAD_COUNT property and specify the number of threads in the following format:

```
-DSATS_THREAD_COUNT=<number of threads>
```

Default is 10.

-DmaxProfilingPoolConnections

Specifies the maximum number of profile mappings that the Secure@Source Service can execute concurrently for one scan job.

The number of profile mappings that a scan job creates depends on the type of profile in the scan. A data profile scan job creates one mapping for each table in the data store. A metadata profile scan job creates one mapping regardless of the number of tables in the data store.

Each mapping uses one unit of the execution pool of the Data Integration Service. If you set the -DmaxProfilingPoolConnections property to the execution pool size, then the mappings from a single data profile scan job might use the total execution pool. To allow multiple scan jobs to run concurrently, minimize the number of execution pool units one scan job can use.

Informatica recommends that you set the value for the -DmaxProfilingPoolConnections property to half or one-third of the value specified in the Maximum Profile Execution Pool Size property of the Data Integration Service.

To specify the maximum number of profile mappings that the Secure@Source Service can execute concurrently, add the -DmaxProfilingPoolConnections property in the following format:

```
-DmaxProfilingPoolConnections=<number of profile mappings>
```

Maximum Heap Size

Maximum JVM heap size. Default is 1024.

Maximum Statements in Cache

Maximum number of cached SQL statements stored in the Secure@Source repository.

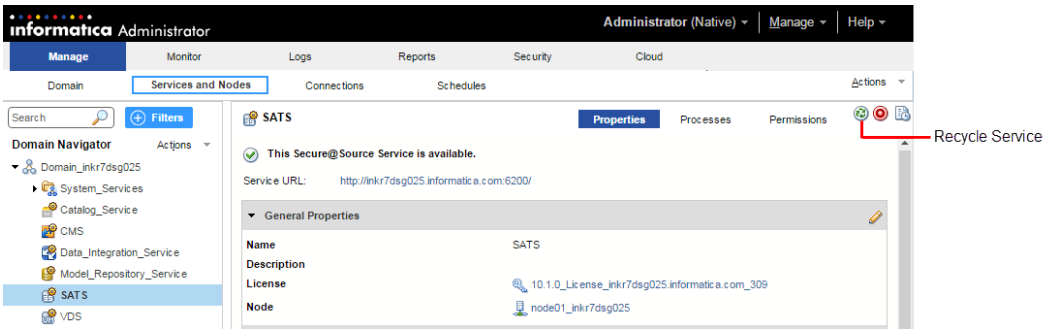
Shared Directory

The directory that Secure@Source shares with the other Informatica services. Default is `<Informatica installation>\server\infa_shared\SecureAtSourceService`.

Step 6c. Recycle the Secure@Source Service

Recycle the Secure@Source Service to apply the updates you made to the service and service process properties. When you recycle the service, the Secure@Source Service is disabled and enabled.
Note: Updates to the Logger Options properties take effect immediately and you do not need to recycle the service if you only updated the Logger Options properties.

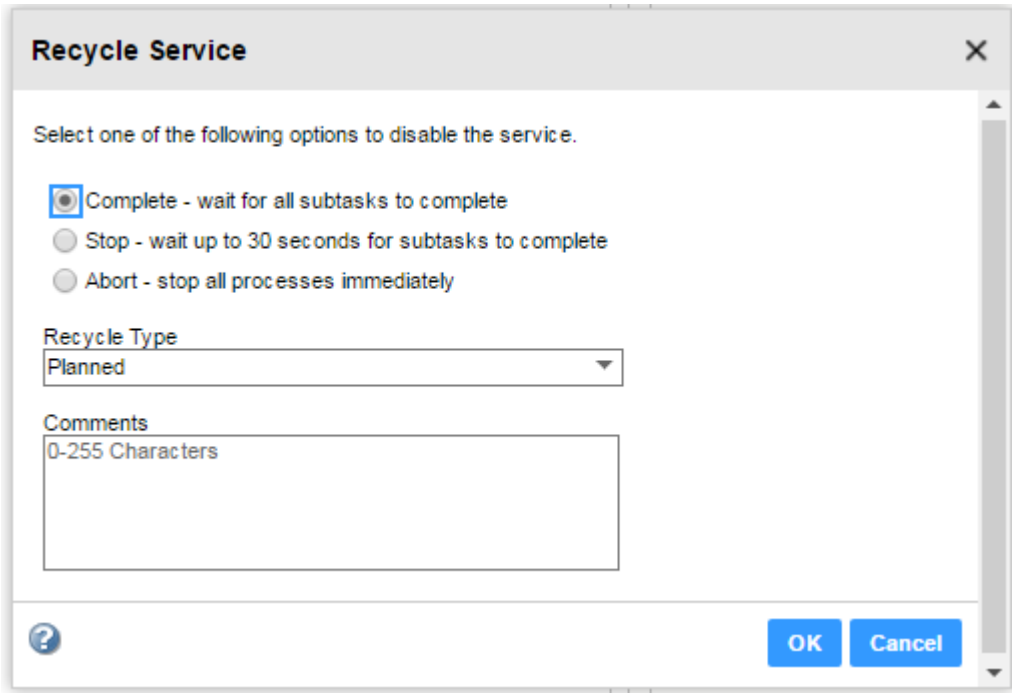
1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. Select the Secure@Source Service in the Domain Navigator pane.
4. Click the Recycle Service icon.



The **Recycle Service** window appears.

5. Select one of the following options:
   - Complete. Allows the jobs to run to completion and then shuts down the service, user activity applications, and Spark components.
   - Stop. Stops the jobs after 30 seconds and then shuts down the service and Spark components.

- Abort. Stops all jobs immediately and then shuts down the service.



6. Optionally, select a recycle type.
7. Optionally, enter a comment such as the reason for recycling the service.
8. Click **OK**.

   The Secure@Source Service shuts down and restarts.

Author